

PUSHPRAJ PANDEY

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LinkedIn GitHub

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering

September 2022 – Ongoing

CGPA: 7.39 /10

TECHNICAL SKILLS

AI & ML: LLMs, Prompt Engineering, RAG Pipelines, Agentic AI, Neural Networks, Deep Learning, Embeddings, Vector Databases, LangChain

Backend & APIs: Python, FastAPI, Node.js, Express.js, REST APIs, Flask

Frontend & Databases: React.js, Next.js, MongoDB, MySQL, Redux

Tools & Cloud: Java, C++, SQL, Git, GCP, AWS, Docker

PROJECTS

SehatStation | *React.js, Node.js, FastAPI, MongoDB, LLM Integration*

September 2024 – October 2024

- Built an end-to-end hospital management platform integrating an LLM-powered AI chatbot for patient triage, appointment scheduling, and real-time OPD queue management across departments
- Designed a RAG-based knowledge retrieval pipeline using semantic embeddings to serve context-aware responses from hospital records, reducing query resolution time by 40%
- Implemented role-based access control, JWT authentication, and RESTful APIs with FastAPI to securely serve 200+ concurrent patient and staff sessions

DriveKaro | *React.js, Node.js, MongoDB, Stripe API, REST APIs*

March 2025 – June 2025

- Engineered a full-stack car rental platform with JWT authentication, live booking engine, and Stripe payment integration, handling 500+ daily API requests reliably
- Built React frontend with Redux state management and optimized backend query performance by 35% through targeted MongoDB indexing and aggregation pipeline tuning
- Deployed Node.js microservices on GCP with automated health checks and environment-based configuration, achieving 99.5% uptime during active booking windows

PennyPilot | *React.js, Node.js, MongoDB, Python, Chart.js*

December 2024 – January 2025

- Developed a full-stack expense management app featuring an NLP-based auto-categorization module using Python that classified 95%+ of transactions without manual input
- Created a responsive analytics dashboard with Chart.js visual embeddings and CRUD APIs, enabling real-time spending pattern insights and wallet balance tracking
- Secured data with role-based access control and RESTful API validation layers, supporting multi-user concurrent sessions without data leakage

Pothole Detection System | *Flutter, Python, OpenCV, Deep Learning*

July 2024 – April 2025

- Engineered a cross-platform mobile application for real-time pothole detection using a convolutional neural network (CNN) model with OpenCV, achieving 88% detection precision on varied road surfaces
- Designed GPS-based geolocation tagging and automated reporting pipeline, centralizing 1,000+ pothole records in a structured database for civic authorities
- Integrated Python backend inference engine with Flutter UI via REST APIs, enabling sub-2s detection latency on standard Android hardware

ACHIEVEMENTS AND CERTIFICATIONS

Certifications: Oracle Data Science Professional, Oracle MySQL HeatWave Implementation, Google: Bits and Bytes of Computer Networking

DSA: Solved 150+ problems on LeetCode (Arrays, Trees, Graphs, Dynamic Programming)

Interests: Generative AI research, Technical Writing, Outdoor Sports

EXTRACURRICULAR ACTIVITIES

Event Organizer – Badminton Co-ordinator, Annual Tech Fest AdvITyA, managed 50+ participants (2024)

Volunteer – NSS (National Service Scheme), community service initiatives (2023–2024)

Sports – Inter-college Badminton Team member